

Fundamental principles and equations governing cellular signal propagation, adaptive gene expression, and biological condensate formation

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We will introduce new chemical dynamics approach for quantitative investigation into complex reaction networks in living cells [1]. In this approach, dynamics of cellular network modules are characterized by reaction-time distributions, rather than the conventional rate constants. Using this framework, we obtain general equations governing cellular signal propagation and adaptive gene expressions, which provides a unified, quantitative explanation of stochastic dynamics of various *E. coli* genes activated by antibiotic stress and transcriptional induction [2]. We will also talk about recently proposed nucleation theories that quantitatively explains growth dynamics and size-distribution of biological condensates [3]. This work sheds light on the thermodynamic origin of stable nuclei formation under supersaturation conditions and provides a unified, quantitative explanation of the nucleus size distribution and size-dependent growth dynamics across various nanoparticles and biological condensates.

References:

- [1] Park et al., *Nature Communications* 9, 297 (2018); Song et al., *PLoS Comp. Biol.* 15, e1007356 (2019); Kang et al., *J. Chem. Phys.* 164, 224127 (2026)
- [2] Kim et al., *Cellular chemical dynamics governing signal transduction and adaptive gene expression: Beyond classical kinetics* bioRxiv (<https://doi.org/10.64898/2026.02.13.705865>); Park et al., *Adv. Sci.* 13, e13823 (2026)
- [3] Kim et al., *Proc. Nat. Acad. Sci. U.S.A.* 122, e2424950122 (2025); Kang et al., *J. Am. Chem. Soc.* 147, 46976 (2025).